

Free Bits: Latency Optimization of Mixed-Precision Quantized Neural Networks on the Edge

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Abstract—Mixed-precision quantization, where a deep neural network's layers are quantized to different precisions, offers the opportunity to optimize the trade-offs between model size, latency, and statistical accuracy beyond what can be achieved with homogeneous-bit-width quantization. To navigate the intractable search space of mixed-precision configurations for a given network, this paper proposes a hybrid search methodology. It consists of a hardware-agnostic differentiable search algorithm followed by a hardware-aware heuristic optimization to find mixed-precision configurations latency-optimized for a specific hardware target. We evaluate our algorithm on MobileNetV1 and MobileNetV2 and deploy the resulting networks on a family of multi-core RISC-V microcontroller platforms with different hardware characteristics. We achieve up to 28.6% reduction of end-to-end latency compared to an 8-bit model at a negligible accuracy drop from a full-precision baseline on the 1000-class ImageNet dataset. We demonstrate speedups relative to an 8-bit baseline, even on systems with no hardware support for sub-byte arithmetic at negligible accuracy drop. Furthermore, we show the superiority of our approach with respect to differentiable search targeting reduced binary operation counts as a proxy for latency.

Index Terms—Edge AI, Mixed-Precision Neural Networks

I. INTRODUCTION

The number of internet of things (IoT) devices deployed is growing rapidly and is projected to reach 19.1 billion by 2025 [1]. To efficiently and accurately process the massive amounts of data collected by IoT sensor nodes under the strict latency and power constraints imposed by IoT applications, the emerging field of Edge AI aims to deploy deep learning (DL) algorithms directly on the edge devices that collect them. Microcontroller units (MCUs) have been a popular target for edge deployment of Deep Neural Networks (DNNs) due to their ubiquity and low cost, and extensive research has been conducted into designing efficient DNN models and techniques to enable inference on MCUs [2], [3]. As the active power consumption of edge nodes is generally dominated by components other than the arithmetic units, the most effective way to decrease the full-system inference energy on a given system is by reducing the inference latency while meeting their tight memory and storage constraints.

A key technique to reduce both memory footprint and inference latency of DNNs is *quantization*, where model parameters and intermediate activations are represented in low-precision formats. 8-bit quantized models generally exhibit

equivalent accuracy to full-precision models. Thanks to Single Instruction Multiple Data (SIMD) instructions, these models can be executed on modern MCU-based system with lower latency and correspondingly reduced energy cost [4]. Quantization to even lower bit-widths has also seen widespread interest [5], [6], [7] and the hardware community has followed suit, proposing low-precision DNN execution engines as well as instruction set architecture (ISA) extensions to accelerate networks quantized to sub-byte precision [8].

However, homogeneous quantization to sub-byte precisions often incurs a non-negligible accuracy penalty. To find the best trade-off between execution latency and statistical accuracy, *mixed-precision quantization* proposes to quantize different parts (usually at the granularity of individual layers) of the network to different precisions. In order to efficiently navigate the intractable search space of precision configurations of a given model, multiple works have applied differentiable neural architecture search (DNAS) to mixed-precision search. These approaches generally rely on a proxy for latency, such as binary operation (BOP) count, to guide the search [9], [10], [11]. By modeling quantization to different precisions in a differentiable manner and adding a regularizer term to the loss to penalize high BOP counts, these algorithms jointly minimize networks' BOP count and task accuracy. The trade-off between operational complexity and accuracy is controlled by the regularization strength.

However, as Figure 1 shows, low BOP counts do not directly translate to reduced execution latency on real hardware platforms, which motivates our work. Depending on the target platform, certain layers even exhibit a higher latency when quantized to lower precisions. One reason for this counter-intuitive behavior is in the hardware implementation of low-precision operations. E.g., the XpulpNN ISA extension used by [8] only supports operands of equal precision - when weight and activation precisions differ, the lower-precision operands must be unpacked to the larger data format. In systems with hierarchically organized memory, tiling effects also shape the precision-latency landscape: larger tiles lead to more efficient execution, with the consequence that some layers exhibit lower latency in sub-8-bit precision on systems without hardware support for sub-8-bit arithmetic (XpulpV2 in Figure 1). Past works have targeted the search for mixed-precision networks for deployment to MCU-class platforms. [12] focuses on reducing the memory/storage footprints, but did not consider inference latency. [13] targets inference energy reduction with a DNAS-based approach to channel-wise

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